Dangers of Electromagnetic Waves

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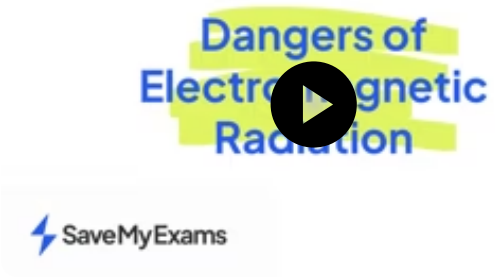

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Dangers of electromagnetic waves

- **Excessive exposure** to electromagnetic radiation has **harmful effects** on people
  - As the frequency of the electromagnetic (EM) waves increases, so does their energy
- Beyond the visible part of the spectrum (ultraviolet waves, X-rays and gamma rays), the energy is large enough to **ionise** atoms
  - The **higher** the **frequency**, the more **ionising** the radiation
- The effects of ionisation also depend on the **type of radiation** and the **size** of the **dose**

Harmful effects of EM radiation

Wave	Danger
Radio	No known danger
Microwaves	Internal heating of body cells

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Ultraviolet	Damage to surface cells and eyes leads to skin cancer and eye conditions
X-rays and gamma rays	Mutation or damage to cells in the body

Harmful effects of excessive microwave exposure

- Certain frequencies of microwaves are absorbed by **water molecules**
- Humans contain a lot of water, so there is a risk of **internal heating** from microwaves
- This might worry some people, but the longer wavelength (lower energy) microwaves used in everyday circumstances are proven to be safe
  - Microwaves used for **communications** (including mobile phones) emit very small amounts of energy which are not known to cause any harm
  - Microwave ovens, on the other hand, emit very large amounts of energy, however, that energy is **prevented** from escaping the oven by the metal walls and metal grid in the glass door

Harmful effects of excessive ultraviolet exposure

- Ultraviolet is similar to visible light, except it is invisible to the human eye and carries a much higher energy
- If eyes are exposed to high levels of UV it can cause **severe** eye damage
  - Good quality sunglasses will absorb ultraviolet, preventing it from entering the eyes
- Ultraviolet can kill cells or cause them to malfunction
  - This can cause **premature ageing**, and diseases such as **skin cancer**
  - Sunscreen absorbs ultraviolet light, preventing it from damaging the skin

Harmful effects of excessive X-ray and gamma-ray exposure

- X-rays and gamma rays are the most ionising types of EM waves
  - They can penetrate the body and cause **internal** damage
  - They can cause the **mutation** of DNA, causing **cancer**

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